

Realtime Scenario-based Interview Questions on MySQL Database Administration, Performance Tuning & Optimization

MySQL Administration (1–150)

1. You are asked to migrate a MySQL database from on-premises to AWS RDS with minimal downtime. How would you approach this?
2. A junior DBA accidentally dropped a table in production. How would you recover it if backups are available?
3. You notice a MySQL server crashing frequently due to corrupted tables. What steps would you take to fix it?
4. The business requires enabling point-in-time recovery (PITR). How would you configure it?
5. A MySQL slave is lagging behind the master by several hours. What steps do you take to troubleshoot?
6. You find that your MySQL error log is filling up disk space. How do you manage and rotate logs effectively?
7. How do you safely upgrade a MySQL instance from version 5.7 to 8.0 in a production environment?
8. Users report frequent “Too many connections” errors. How would you handle this issue?
9. You need to enforce data encryption at rest for MySQL. What options would you configure?
10. A replication slave has stopped with a duplicate key error. How would you resolve it without restarting replication from scratch?
11. Your company requires multi-region failover for MySQL. Which architecture would you recommend?
12. A large import operation is slowing down MySQL performance. What optimizations would you apply?
13. A MySQL instance is running out of disk space. What quick actions would you take?
14. You are tasked with setting up MySQL Group Replication for high availability. What are the key considerations?
15. How do you manage user privileges across multiple environments securely?
16. What’s your approach to rotating database user passwords without downtime?
17. You discover unauthorized queries being executed on MySQL. How do you investigate and secure the instance?
18. You need to migrate MySQL tables from MyISAM to InnoDB. How do you plan this?
19. How do you automate schema changes across multiple MySQL servers?
20. A query suddenly starts causing deadlocks. How would you resolve and prevent this?
21. Your application team requests audit logging of all data modifications. How would you implement it?
22. How do you configure MySQL for high write throughput in a financial system?
23. After enabling SSL, clients fail to connect. How would you debug and fix this?
24. You need to enforce row-level security in MySQL. How would you design this?
25. During a failover test, your MySQL replication broke. What’s your recovery plan?
26. Your production database is too big to back up with mysqldump. What alternatives would you use?
27. A MySQL cluster node is repeatedly crashing. How would you perform root cause analysis?
28. How do you handle schema versioning in MySQL when multiple teams are deploying changes?
29. You detect memory leaks in MySQL. How do you confirm and fix this?
30. An application requires zero downtime schema migrations. What strategy would you use?
31. You notice slow restart times after crashes due to long InnoDB recovery. How do you optimize recovery?
32. MySQL is showing “table is marked as crashed and should be repaired.” How do you fix this?
33. Your backups are taking too long to complete. How do you optimize backup processes?
34. How do you perform rolling upgrades for MySQL cluster nodes?
35. A user accidentally truncated a table. How do you restore only that table?
36. MySQL is running in a container environment. What special considerations do you take?
37. You are asked to design MySQL for GDPR compliance. What features/configurations would you enable?
38. You find a high number of aborted connections. How do you investigate the cause?

39. How do you detect and prevent MySQL data corruption proactively?
40. A developer has direct write access to production. How do you enforce database governance?
41. Your replication topology requires adding a new slave without downtime. How do you proceed?
42. MySQL server is showing high CPU usage even during low query load. How do you troubleshoot?
43. You need to archive old data to keep table sizes manageable. How do you implement this?
44. MySQL binlogs are growing uncontrollably. What do you check?
45. You are asked to configure MySQL with LDAP authentication. How do you achieve it?
46. You need to enable data masking for sensitive fields. How would you approach this?
47. MySQL slow query log is not rotating properly. How do you manage this?
48. You are tasked with setting up MySQL multi-source replication. How would you configure it?
49. How do you detect and resolve foreign key constraint violations?
50. Your replication slave shows "Relay log read failure." How do you fix it?
51. During a backup window, users report query slowness. How do you adjust backup strategies to reduce impact?
52. You have a 1TB MySQL database. What's the safest way to back it up daily without long downtime?
53. How do you implement incremental backups in MySQL, and when would you prefer them over full backups?
54. If your mysqldump backup is taking too long, what real-time optimizations would you apply?
55. What steps do you follow to restore a backup into a standby server for disaster recovery?
56. You need point-in-time recovery (PITR). How do you configure MySQL binary logs to support it?
57. During a failover test, you observe replication lag on the standby. How do you troubleshoot and fix it?
58. What are the differences between cold backup, hot backup, and logical backup in MySQL, and when to use each?
59. How do you validate a MySQL backup for corruption before restoring?
60. A backup restoration fails with "invalid format." What steps do you take to recover data?
61. How do you handle user management in MySQL across multiple environments (Dev, QA, Prod) without drift?
62. What security best practices would you follow when creating MySQL user accounts?
63. A user is complaining about "Access denied" errors. How do you debug GRANT issues in MySQL?
64. How do you enforce password rotation policies for MySQL user accounts?
65. You need to provide temporary read-only access to a consultant. How would you implement it securely?
66. How do you audit who dropped a table in MySQL?
67. What methods are available for enabling query auditing in MySQL?
68. How do you configure MySQL Enterprise Audit plugin vs. open-source alternatives (Percona Audit Log)?
69. A developer accidentally granted SUPER privilege to an application user. How do you mitigate this risk?
70. How do you safely revoke user privileges without breaking application functionality?
71. You are planning an in-place MySQL upgrade from 5.7 to 8.0. What pre-checks would you perform?
72. During upgrade, you encounter incompatible character sets. How do you handle it?
73. How do you perform a rolling upgrade of a MySQL replication cluster with minimal downtime?
74. Your production upgrade fails mid-way. What rollback strategy would you follow?
75. How do you test schema and application compatibility before upgrading MySQL?
76. What's the difference between mysql_upgrade and mysqlcheck during an upgrade process?
77. How do you upgrade MySQL in a Galera Cluster environment?
78. What risks do you face when skipping intermediate versions during upgrade?
79. How do you ensure replication compatibility when upgrading master before replicas?
80. How do you plan upgrades in an environment with multiple MySQL engines (InnoDB, MyISAM, etc.)?
81. What monitoring metrics do you enable to track MySQL server health daily?
82. How do you set up real-time monitoring for replication lag?

83. What alerts would you configure in Prometheus/Grafana for MySQL?
84. A sudden CPU spike is reported on MySQL server. How do you investigate live?
85. How do you differentiate between MySQL performance issues vs. OS-level issues?
86. Which logs do you analyze first when MySQL becomes unresponsive?
87. How do you monitor long-running queries in MySQL?
88. A table is locked for too long. How do you identify the session holding the lock?
89. How do you configure the slow query log for targeted query analysis?
90. How do you set up MySQL Enterprise Monitor vs. open-source monitoring (Percona Monitoring and Management)?
91. How do you plan MySQL failover in a master-replica setup?
92. What are the differences between manual failover, semi-automatic, and automatic failover?
93. How do you test failover regularly without disrupting production traffic?
94. A failover resulted in split-brain. How do you prevent it in MySQL replication setups?
95. How do you configure MySQL with HAProxy for automatic failover?
96. What's your strategy to handle replication lag during failover?
97. How do you perform controlled failover during planned maintenance?
98. After failover, the new primary shows missing transactions. How do you sync?
99. What's the difference between MySQL replication failover vs. Group Replication failover?
100. How do you ensure application reconnection after a MySQL failover event?
101. How do you manage schema changes in large production databases with minimal downtime?
102. What online schema change tools do you use (pt-online-schema-change, gh-ost)?
103. How do you rollback a failed schema change in production?
104. How do you test schema changes for replication safety?
105. A schema migration causes replication to break. How do you fix it?
106. How do you handle adding new indexes without locking the table for long?
107. How do you automate schema deployments across multiple environments?
108. How do you manage foreign key constraints in schema changes during peak traffic?
109. How do you detect unused indexes before dropping them?
110. How do you handle character set conversions during schema upgrades?
111. How do you handle database growth monitoring and capacity planning?
112. How do you archive old data without affecting active workloads?
113. How do you partition large tables to improve manageability?
114. How do you purge binary logs safely without breaking replication?
115. How do you shrink an oversized InnoDB log file?
116. How do you handle data purging in a GDPR-compliant environment?
117. How do you forecast storage requirements for next year?
118. How do you compress backups for faster transfer?
119. How do you split large dump files for parallel restore?
120. How do you plan sharding in MySQL for extreme growth?
121. How do you configure MySQL error logging for better troubleshooting?
122. What's your approach to rotate MySQL logs automatically?
123. How do you collect audit trails of login and failed login attempts?
124. How do you track schema change history in MySQL?
125. What tools do you use for log aggregation from multiple MySQL servers?
126. How do you enable query sampling for auditing in MySQL?
127. How do you implement row-level auditing without impacting performance?

- 128. How do you secure audit logs from tampering?
- 129. How do you detect suspicious queries from logs in real-time?
- 130. How do you automate audit report generation for compliance?
- 131. How do you configure replication for DR (Disaster Recovery)?
- 132. What's your approach to testing DR site failover?
- 133. How do you reduce RPO and RTO in MySQL disaster recovery?
- 134. How do you maintain backups in geographically distributed sites?
- 135. How do you encrypt backups in transit and at rest?
- 136. How do you handle backup failures in a DR drill?
- 137. What's your process to rebuild a failed replica quickly?
- 138. How do you keep replicas consistent after network outage?
- 139. How do you manage DR for multi-tenant MySQL clusters?
- 140. How do you test PITR (Point-In-Time Recovery) during DR drills?

- 141. How do you handle user onboarding automation in MySQL?
- 142. How do you integrate MySQL authentication with LDAP/AD?
- 143. How do you enforce SSL/TLS connections for MySQL clients?
- 144. How do you set up MySQL roles for easier privilege management?
- 145. How do you detect and block SQL injection attempts in MySQL?
- 146. How do you configure firewall rules to secure MySQL?
- 147. How do you perform security hardening for MySQL post-installation?
- 148. How do you rotate MySQL certificates before they expire?
- 149. How do you manage secrets (passwords) securely for MySQL connections?
- 150. How do you handle GDPR/PII compliance in MySQL auditing and logging?

Performance Tuning (151–350)

- 151. An application team complains of query timeouts. How do you approach tuning?
- 152. MySQL shows high I/O wait. How do you tune it?
- 153. How would you analyze a slow query using EXPLAIN?
- 154. What's your approach if queries frequently trigger temporary tables on disk?
- 155. How do you optimize a MySQL instance with high concurrency?
- 156. You notice excessive row scans in queries. How do you fix this?
- 157. MySQL shows frequent full table scans despite indexes. What's the cause?
- 158. The query cache is causing contention. What tuning changes would you apply?
- 159. How do you tune MySQL buffer pool for optimal performance?
- 160. You are asked to tune MySQL for read-heavy workloads. What parameters do you adjust?
- 161. Write-heavy transactions are locking rows for too long. How do you optimize?
- 162. You see high numbers of "waiting for table metadata lock" errors. What's your action plan?
- 163. A query using JOINS across multiple large tables is very slow. How would you optimize it?
- 164. How do you troubleshoot excessive temporary table creation?
- 165. You find replication lag due to long-running queries. How do you tune queries to avoid lag?
- 166. A query is using an index but still slow. What might be the reasons?
- 167. How do you optimize a MySQL database for SSD storage?
- 168. What's your approach to analyzing performance using Performance Schema?
- 169. MySQL is running out of connections. How do you balance connection pooling?

170. How do you identify unused indexes for removal?
171. You find queries frequently blocked by foreign key checks. How do you optimize schema design?
172. InnoDB buffer pool hit ratio is very low. What's your tuning approach?
173. How do you optimize SELECT queries with ORDER BY and LIMIT on large datasets?
174. You are asked to reduce query response times under 200 ms. What steps do you take?
175. MySQL shows a lot of context switching in OS metrics. How do you tune?
176. You observe row-level locks escalating. How do you prevent performance bottlenecks?
177. A query using subqueries is too slow. How would you rewrite it?
178. How do you optimize table partitioning in MySQL?
179. Queries are running fine but API latency is high. How do you prove/disprove MySQL is the cause?
180. How do you use indexes to speed up LIKE queries?
181. Your application requires reporting queries. How do you optimize read performance without affecting OLTP?
182. MySQL's default isolation level is causing performance issues. How do you tune it?
183. High redo log writes are slowing performance. How do you tune redo logs?
184. You see frequent "Row lock wait timeout exceeded" errors. How do you optimize queries?
185. Queries using UNION are slow. How do you optimize them?
186. You detect a high percentage of queries without indexes. What's your strategy to fix them?
187. How do you tune a MySQL server for batch inserts?
188. A query against a large table runs fast sometimes and slow at other times. How do you investigate?
189. How do you optimize replication performance in write-heavy environments?
190. You find MySQL is spending time on sorting. How do you tune sort buffer settings?
191. How do you profile queries to detect bottlenecks?
192. MySQL thread pool is maxed out. How do you tune thread usage?
193. You see MySQL creating too many open files. How do you tune file descriptors?
194. High lock contention is slowing down performance. How do you address it?
195. Queries with OR conditions are slow. How do you optimize?
196. How do you tune MySQL for hybrid workloads (OLTP + OLAP)?
197. You notice a lot of dead tuples in InnoDB. How do you reclaim space?
198. Queries with GROUP BY are slow. How do you optimize them?
199. MySQL optimizer is not picking the best index. How do you force or guide it?
200. You see increasing query execution times over weeks. What's your approach to performance trending?

◆ MySQL Performance Tuning & Optimization Scenarios (201–350)

201. A query is running slow even though it uses indexes. How do you debug index selectivity issues?
202. How do you identify if a query is performing a full table scan despite having an index?
203. A query using OR is slow. How do you optimize queries with multiple OR conditions?
204. How do you optimize queries with IN vs. EXISTS in MySQL?
205. How do you detect redundant joins in complex queries?
206. A query with a LIKE '%word%' condition is slow. What index strategies can you apply?
207. How do you optimize subqueries that perform poorly compared to joins?
208. How do you optimize GROUP BY queries with millions of rows?

209. What's your approach to handling SELECT * queries in production?
210. How do you optimize ORDER BY with LIMIT queries for pagination?
211. How do you identify missing indexes from slow query logs?
212. A query with composite index (col1, col2, col3) is slow. How do you validate index usage?
213. How do you decide between covering indexes vs. normal indexes?
214. What's your approach to handle duplicate indexes in large schemas?
215. How do you optimize queries involving text search?
216. How do you evaluate if an index is bloated or fragmented?
217. How do you detect unused indexes in production?
218. A query uses multiple indexes but optimizer picks the wrong one. How do you fix it?
219. How do you handle too many indexes slowing down write operations?
220. How do you apply partial indexing strategies in MySQL 8?
221. How do you tune schema design for OLTP workloads with high concurrency?
222. How do you decide between normalization vs. denormalization?
223. A table with wide VARCHAR columns is slowing queries. How do you optimize schema storage?
224. How do you handle large blob/text columns in schema design?
225. How do you optimize surrogate keys vs. natural keys in schema?
226. How do you detect schema anomalies that cause inefficient joins?
227. What's your approach for designing schemas for time-series data?
228. How do you decide partitioning strategies in schema design?
229. How do you ensure schema design supports multi-tenant workloads?
230. How do you manage schema evolution without breaking apps?
231. How do you monitor buffer pool hit ratio in MySQL?
232. A workload shows frequent disk IO despite high buffer pool size. How do you debug it?
233. How do you decide the right innodb_buffer_pool_size for a production system?
234. How do you handle buffer pool sizing when running multiple MySQL instances?
235. How do you tune innodb_log_buffer_size for heavy write workloads?
236. How do you optimize flush policy (innodb_flush_log_at_trx_commit) for durability vs. performance?
237. How do you identify buffer pool contention in high-concurrency workloads?
238. How do you warm up buffer pool after restart for faster performance?
239. How do you check which tables consume most buffer pool memory?
240. How do you monitor buffer pool dirty pages and tune checkpointing?
241. How do you tune thread concurrency in MySQL for OLTP workloads?
242. What's your approach to configure innodb_thread_concurrency?
243. How do you debug thread contention on InnoDB row locks?
244. How do you handle excessive threads waiting in processlist?
245. How do you monitor thread scheduling in MySQL under peak load?
246. How do you optimize background threads like purge and checkpoint threads?
247. How do you handle performance degradation due to too many idle connections?
248. How do you tune thread_cache_size for high connection churn?
249. How do you reduce context switching in thread-heavy workloads?
250. How do you decide between thread-per-connection vs. thread pool plugin?
251. How do you balance workloads across read replicas?
252. What strategies do you use to split read and write queries?

253. How do you configure HAProxy/ProxySQL for workload balancing in MySQL?
254. How do you prevent read-after-write inconsistencies in workload balancing?
255. How do you handle hot rows being accessed by multiple replicas?
256. How do you configure weighted load balancing across replicas?
257. How do you detect replica lag and redirect traffic?
258. How do you simulate load balancing failure scenarios?
259. How do you manage workload balancing in a sharded MySQL environment?
260. How do you integrate application-side load balancing with MySQL proxies?
261. How do you force the optimizer to use a specific index with hints?
262. How do you disable index usage temporarily for testing?
263. How do you apply STRAIGHT_JOIN to optimize join order?
264. When would you use optimizer hints like USE INDEX or IGNORE INDEX?
265. How do you verify if optimizer hints are actually applied?
266. How do you handle optimizer choosing suboptimal execution plans?
267. How do you use MAX_EXECUTION_TIME hint to avoid runaway queries?
268. How do you force a join order in a query execution plan?
269. How do you use hints in partitioned tables for performance?
270. How do you prevent misuse of optimizer hints in production code?
271. How do you analyze EXPLAIN output for queries?
272. How do you interpret "Using temporary" in EXPLAIN plans?
273. How do you detect when filesort is being used unnecessarily?
274. How do you compare execution plans before and after schema/index changes?
275. How do you use ANALYZE FORMAT=JSON to debug optimizer choices?
276. How do you detect index condition pushdown usage in execution plans?
277. How do you identify range scans vs. index scans in EXPLAIN?
278. How do you check if partition pruning is effective in execution plans?
279. How do you measure actual query execution time vs. estimated cost?
280. How do you cache execution plans effectively in MySQL?
281. How do you handle queries that generate too many temporary tables?
282. How do you optimize queries that spill to disk temp tables?
283. How do you tune tmp_table_size and max_heap_table_size?
284. How do you detect queries causing table cache misses?
285. How do you tune table_open_cache for high-concurrency workloads?
286. How do you reduce contention in metadata locking?
287. How do you optimize queries with DISTINCT and GROUP BY?
288. How do you analyze SHOW PROFILE to identify performance bottlenecks?
289. How do you configure query cache (legacy) vs. application caching?
290. How do you reduce sort buffer usage in ORDER BY heavy workloads?
291. How do you detect memory leaks in MySQL queries?
292. How do you optimize join buffers for star schema queries?
293. How do you tune key_buffer_size for MyISAM workloads?
294. How do you optimize redo log configuration for high writes?
295. How do you configure doublewrite buffer for performance?
296. How do you detect deadlocks using InnoDB status?
297. How do you analyze performance impact of foreign keys?

- 298.** How do you avoid locking issues in bulk updates?
- 299.** How do you optimize DELETE with millions of rows?
- 300.** How do you analyze performance differences between DELETE vs. TRUNCATE?

- 301.** How do you detect hotspot rows causing contention?
- 302.** How do you partition large tables to reduce hotspots?
- 303.** How do you tune row-level locking in InnoDB?
- 304.** How do you analyze lock waits with Performance Schema?
- 305.** How do you resolve deadlocks between long-running transactions?
- 306.** How do you tune isolation levels for performance vs. consistency?
- 307.** How do you optimize retry logic for deadlock victims?
- 308.** How do you handle phantom reads in repeatable read isolation?
- 309.** How do you analyze row locking vs. gap locking performance?
- 310.** How do you reduce lock escalation in MySQL?

- 311.** How do you optimize SELECT COUNT(*) queries on huge tables?
- 312.** How do you speed up analytical queries in MySQL without OLAP features?
- 313.** How do you design summary tables for reporting workloads?
- 314.** How do you cache aggregated results effectively?
- 315.** How do you tune MySQL partition pruning for analytics?
- 316.** How do you detect queries that bypass partitioning benefits?
- 317.** How do you optimize star joins in data warehouse schemas?
- 318.** How do you handle rollup queries with GROUPING SETS in MySQL 8?
- 319.** How do you analyze histogram statistics impact on optimizer?
- 320.** How do you tune MySQL histograms for skewed data distributions?

- 321.** How do you configure workload classes in ProxySQL for query prioritization?
- 322.** How do you detect resource-hungry queries using Performance Schema?
- 323.** How do you optimize ad-hoc query workloads without hurting OLTP?
- 324.** How do you use query rewrite plugins for optimization?
- 325.** How do you enforce query timeouts for bad queries?
- 326.** How do you identify and kill runaway queries in real time?
- 327.** How do you configure max connections vs. thread pool for workload balance?
- 328.** How do you rate-limit expensive queries in ProxySQL?
- 329.** How do you detect high-frequency queries degrading performance?
- 330.** How do you use connection multiplexing to optimize workloads?

- 331.** How do you analyze memory consumption of queries in Performance Schema?
- 332.** How do you optimize buffer pool allocation across multiple schemas?
- 333.** How do you detect buffer pool thrashing in workloads?
- 334.** How do you optimize redo log size for heavy transactions?
- 335.** How do you use persistent optimizer statistics across restarts?
- 336.** How do you tune replication filtering for performance?
- 337.** How do you detect queries causing replication lag?
- 338.** How do you optimize large transaction replication?
- 339.** How do you optimize row-based vs. statement-based replication?
- 340.** How do you configure parallel replication for heavy workloads?

- 341. How do you analyze performance schema events to tune queries?
- 342. How do you monitor wait events in MySQL for tuning?
- 343. How do you detect I/O bottlenecks at query level?
- 344. How do you tune redo/undo tablespaces for high-performance workloads?
- 345. How do you measure contention on undo logs?
- 346. How do you tune purge operations in InnoDB?
- 347. How do you debug excessive background flushing in MySQL?
- 348. How do you tune adaptive hash index performance?
- 349. How do you detect optimizer misestimates using histograms?
- 350. How do you validate query tuning improvements using benchmark tests?

Optimization & Troubleshooting (351–500)

- 351. How do you reduce downtime during schema changes?
- 352. You need to optimize a large reporting query without affecting OLTP workload. What's your approach?
- 353. How would you design indexes for a multi-tenant database?
- 354. A batch process is taking 10 hours. How do you reduce it to 1 hour?
- 355. You're asked to optimize MySQL replication across WAN links. What tuning would you apply?
- 356. A customer complains about inconsistent query performance. How do you identify root cause?
- 357. How do you optimize MySQL startup and shutdown times?
- 358. MySQL is swapping heavily. What memory tuning would you do?
- 359. You're tasked with optimizing JSON queries in MySQL. How do you approach it?
- 360. How do you optimize MySQL on Kubernetes clusters?
- 361. MySQL writes are slower after enabling foreign keys. How do you optimize?
- 362. How would you reduce binlog disk usage without affecting replication?
- 363. How do you implement query plan stability in MySQL?
- 364. A table has billions of rows and needs frequent lookups. How do you optimize it?
- 365. You're asked to reduce replication lag from 10 minutes to under 10 seconds. How do you achieve it?
- 366. MySQL slow log shows thousands of queries per second. How do you prioritize optimization?
- 367. How do you optimize MySQL schema for IoT workloads?
- 368. MySQL full-text search queries are slow. How do you optimize them?
- 369. You need to scale out reads significantly. How do you design the architecture?
- 370. MySQL backups are locking tables during peak hours. How do you avoid this?
- 371. How do you optimize MySQL queries for geo-spatial data?
- 372. A partitioned table is slower than expected. How do you tune partitions?
- 373. How do you prevent regression in query performance after upgrades?
- 374. You find MySQL optimizer statistics are outdated. How do you refresh and tune stats?
- 375. How would you design indexes for a table with frequent writes?
- 376. A query is using an index merge plan but running slow. How do you optimize?
- 377. How do you tune for workloads with millions of small transactions?
- 378. MySQL connections keep getting dropped. How do you optimize networking parameters?
- 379. You are asked to optimize queries with CTEs. What's your approach?
- 380. MySQL CPU utilization spikes at specific times. How do you optimize workload distribution?
- 381. You notice table fragmentation. How do you defragment tables?
- 382. How do you optimize MySQL for cloud-native deployments?

383. MySQL writes are bottlenecked by redo logs. How do you tune?
384. How do you implement connection pooling effectively?
385. MySQL replication is causing write amplification. How do you optimize?
386. How do you optimize purge operations in InnoDB?
387. A query's execution plan changes frequently. How do you stabilize it?
388. How do you tune MySQL caching layers?
389. You need to optimize cross-database joins. How would you approach this?
390. MySQL shows high mutex contention. How do you optimize concurrency?
391. A stored procedure is running very slowly. How do you optimize it?
392. You need to optimize ETL processes running in MySQL. What's your strategy?
393. How do you optimize high cardinality indexes?
394. Queries using HAVING are slow. How do you optimize?
395. How would you tune replication when network latency is unpredictable?
396. MySQL inserts are slow due to triggers. How do you optimize triggers?
397. You detect redundant indexes slowing down writes. How do you handle them?
398. You need to optimize sharding in MySQL. How would you approach it?
399. MySQL full table scans are unavoidable for some queries. How do you reduce impact?
400. How do you optimize backup and restore times for multi-terabyte databases?
401. You need to reduce MySQL query latency under heavy workloads. What's your method?
402. How do you design MySQL indexes for star schema reporting?
403. MySQL shows high redo log contention. How do you resolve it?
404. You need to tune queries involving window functions. What's your process?
405. How do you prevent optimizer from picking inefficient plans?
406. How do you optimize table compression in InnoDB?
407. Queries with DISTINCT are slow. How do you optimize?
408. You need to optimize query retries during failovers. How do you handle it?
409. How do you optimize write-intensive OLTP workloads?
410. How do you tune auto-increment columns for scalability?
411. A replication slave keeps falling behind during peak hours. How do you optimize it?
412. How do you minimize downtime during partition maintenance?
413. You're tasked to tune MySQL for financial transactions. What's your focus?
414. A query with JOIN + GROUP BY is extremely slow. How do you rewrite it?
415. How do you optimize rollback segment usage in MySQL?
416. How do you design indexes for time-series data in MySQL?
417. You need to optimize application-side caching with MySQL. How do you integrate it?
418. Queries using OR are slow. How do you restructure them?
419. How do you optimize replication with parallel workers?
420. A MySQL upgrade caused regression in query speed. How do you troubleshoot?
421. You need to reduce load on primary by scaling reads. How do you achieve this?
422. How do you optimize MySQL for mixed OLTP + OLAP workloads?
423. MySQL buffer pool size is maxed out. How do you optimize further?
424. You're tasked with optimizing MySQL performance in Azure/AWS/GCP. What's your approach?
425. A MySQL cluster has hotspots on certain nodes. How do you rebalance?

426. Queries involving ENUM fields are slow. How do you optimize schema design?
427. How do you optimize temp table usage in queries?
428. How do you optimize stored functions?
429. Queries with correlated subqueries are too slow. How do you optimize them?
430. You are asked to reduce MySQL downtime during failovers. How do you achieve it?
431. MySQL binary logs are causing storage pressure. How do you tune retention?
432. How do you optimize large table ALTER operations?
433. Queries with UNION ALL are slow. How do you optimize them?
434. You need to tune MySQL under sudden traffic spikes. What's your strategy?
435. How do you detect and optimize MySQL hot rows?
436. MySQL is slow after a schema migration. How do you troubleshoot?
437. How do you optimize sharded queries?
438. You notice excessive thread contention. How do you tune MySQL thread pool?
439. How do you optimize database access from microservices?
440. MySQL latency spikes during backups. How do you optimize?
441. How do you tune redo log flushing frequency?
442. How do you optimize MySQL semi-synchronous replication?
443. MySQL partition pruning isn't working. How do you fix it?
444. How do you optimize batch deletes?
445. How do you tune MySQL for large text/blob workloads?
446. Queries involving CROSS JOIN are slow. How do you optimize?
447. You need to reduce storage costs without affecting performance. What's your approach?
448. MySQL optimizer is picking wrong join order. How do you fix it?
449. How do you optimize purge threads in InnoDB?
450. You need to reduce query response time by 50%. What's your plan?
451. MySQL is heavily fragmented. How do you reclaim performance?
452. How do you tune MySQL checkpointing?
453. Queries using IN lists with thousands of values are slow. How do you optimize?
454. MySQL fails under sudden write bursts. How do you optimize?
455. How do you tune MySQL caching for SELECT-heavy workloads?
456. Queries with regex filters are slow. How do you optimize?
457. How do you tune MySQL redo log buffer size?
458. Queries with multi-column indexes are slow. How do you tune them?
459. MySQL read replicas are inconsistent. How do you fix and tune them?
460. How do you tune MySQL's query optimizer cost model?
461. How do you optimize MySQL for blockchain/ledger use cases?
462. Queries using NOT IN are slow. How do you optimize?
463. MySQL thread concurrency is hitting OS limits. How do you optimize?
464. How do you optimize stored procedure loops?
465. You need to optimize MySQL replication failover time. How do you configure it?
466. MySQL logs show frequent deadlocks. How do you optimize schema and queries?
467. How do you optimize MySQL with external caching systems (Redis, Memcached)?
468. You need to scale MySQL to 10x current workload. What's your plan?

469. How do you optimize global temporary tables?
470. MySQL optimizer doesn't use covering indexes. How do you enforce it?
471. Queries with LEFT JOIN are slow. How do you optimize?
472. You need to tune MySQL optimizer for large data warehouses. How do you proceed?
473. How do you optimize partitioned indexes?
474. MySQL latency is unpredictable. How do you stabilize performance?
475. How do you tune redo log file sizes?
476. Queries with EXISTS are slow. How do you optimize?
477. How do you tune MySQL IO scheduler?
478. MySQL redo log flushing slows down writes. How do you optimize?
479. How do you optimize MySQL query plan caching?
480. Queries using CASE expressions are slow. How do you optimize?
481. How do you tune MySQL adaptive hash index?
482. MySQL optimizer fails with complex queries. How do you manually optimize?
483. How do you optimize bulk updates?
484. Queries using Cartesian joins are killing performance. How do you rewrite them?
485. How do you optimize MySQL disk I/O patterns?
486. You notice uneven replication load. How do you balance it?
487. How do you optimize MySQL redo log write latency?
488. MySQL optimizer ignores some indexes. How do you troubleshoot?
489. How do you optimize MySQL replication semi-sync performance?
490. You need to tune MySQL buffer pool for mixed workloads. How do you proceed?
491. MySQL background threads are overloaded. How do you optimize?
492. How do you tune MySQL durability vs performance trade-offs?
493. Queries with nested views are slow. How do you optimize?
494. MySQL replication writes are delayed. How do you tune relay logs?
495. How do you optimize MySQL caching for OLAP queries?
496. Queries with JOIN + DISTINCT are slow. How do you tune them?
497. How do you optimize background purge operations in InnoDB?
498. MySQL optimizer overestimates row counts. How do you tune statistics?
499. How do you optimize MySQL in a serverless environment?
500. You are tasked to bring query latency down from 500ms to <50ms. How would you approach end-to-end optimization?